CHAPTER 13 / SUPPLEMENTARY MATERIAL

Cyber–Physical Security and Mission Assurance for Unmanned Aerial Vehicles

UAS Intelligence, Surveillance, and Reconnaissance Security

Secure Payload Integration for ISR Platforms



Figure S13.1. Secure payload integration for intelligence, surveillance, and reconnaissance (ISR) platforms. A securely mounted and authenticated electro-optical/infrared (EO/IR) payload connects to the onboard processor through an avionics interface with isolated power and data paths. Firmware verification, encrypted transmission, and data integrity and provenance checks protect the imagery from capture to ground-station review.

EO/IR, Radar, LiDAR, and Hyperspectral Sensor Security

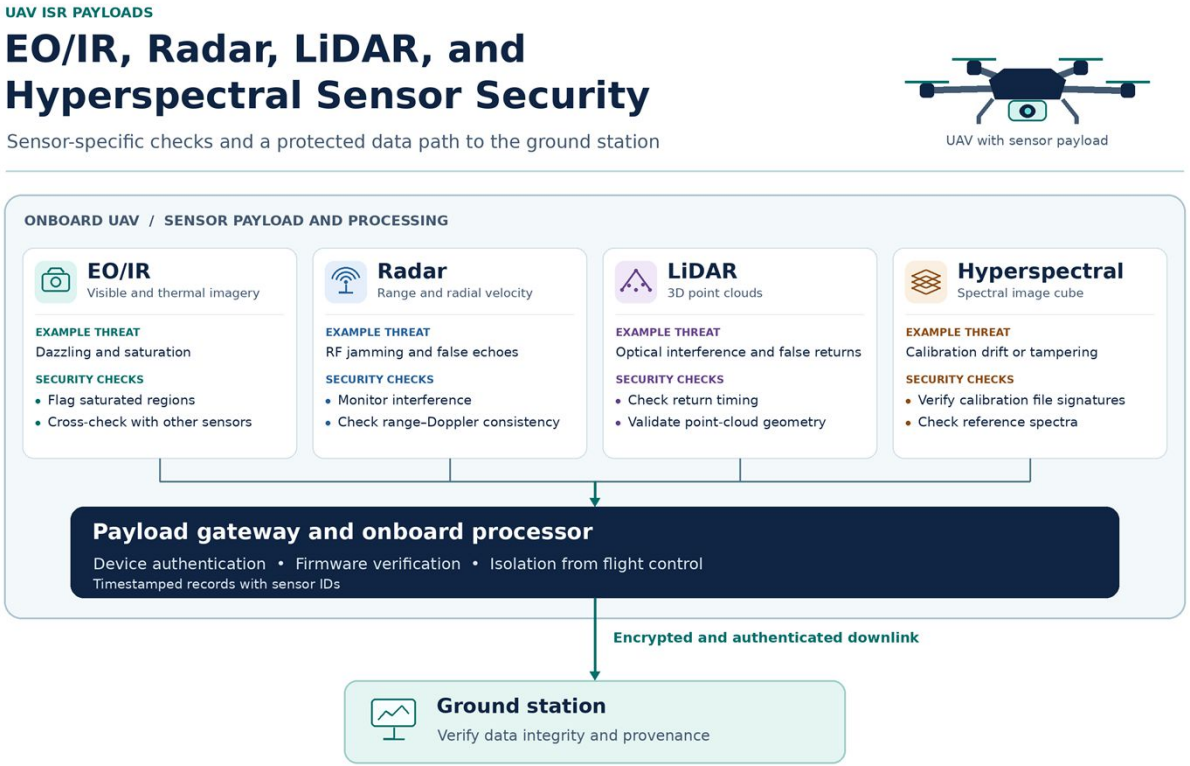


Figure S13.2. Sensor security for UAV-based intelligence, surveillance, and reconnaissance (ISR). The schematic pairs EO/IR, radar, LiDAR, and hyperspectral sensors with example threats and validation checks. A shared onboard gateway authenticates devices, verifies firmware, and isolates the payload from flight control. An encrypted and authenticated downlink carries sensor data to the ground station for integrity and provenance checks.

Data Integrity and Trustworthiness in ISR Pipelines

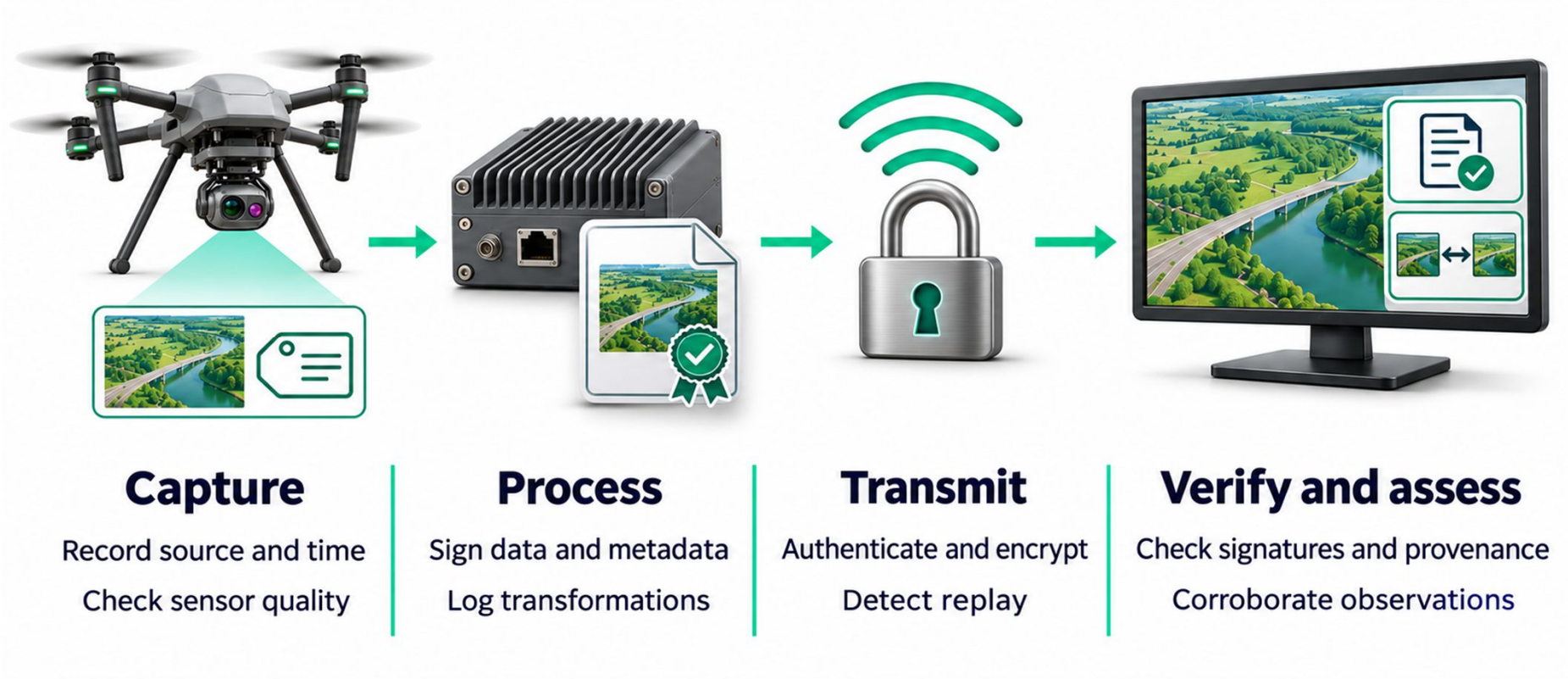


Figure S13.3. Data integrity and trustworthiness in UAV intelligence, surveillance, and reconnaissance (ISR) pipelines. The UAV records the source and time of capture, signs the data and metadata, and logs processing steps before sending the record over an authenticated, encrypted link. At the ground station, signature and provenance checks verify the digital record, while independent observations help assess the reliability of what the sensor captured.

Secure Storage, Transmission, and Processing of ISR Data



Figure S13.4. Secure storage, transmission, and processing of intelligence, surveillance, and reconnaissance (ISR) data. The UAV encrypts stored sensor data and restricts access to encryption keys. An authenticated, encrypted wireless link protects data in transit. At the ground station, isolated workloads process the imagery, and audit logs record access and changes.

AI-Assisted ISR and Vulnerabilities of Intelligent Perception

Mission scenario

A UAV uses electro-optical (EO) and thermal infrared (IR) cameras to survey an industrial site on an intelligence, surveillance, and reconnaissance (ISR) mission.

Onboard detection An AI detector identifies vehicles in sensor imagery.

Track fusion The fusion module combines detections into vehicle tracks.

Ground-station review The ground station scores each track and decides whether to accept it or send it for operator review.

Compare a weighted score with a rule that also enforces minimum input quality.

AI-Assisted ISR and Vulnerabilities of Intelligent Perception

Weighted trust model

$$\tau(t) = \alpha S(t) + \beta M(t) + \gamma F(t) + \delta D(t) + \eta C(t)$$

All inputs lie in [0, 1]. Weights are nonnegative and sum to 1.

Input	Weight	What the input represents
S Sensor quality	0.25	Quality of the sensor observations
M Model confidence	0.25	Confidence reported by the detector
F Fusion agreement	0.20	Apparent agreement among fused inputs
D Metadata validity	0.15	Validity of timestamps and source identifiers
C Context consistency	0.15	Agreement with independent evidence and mission context

AI-Assisted ISR and Vulnerabilities of Intelligent Perception

Simulated observations

Case	Sensor S	Model M	Fusion F	Metadata D	Context C
A. Nominal	0.90	0.92	0.88	1.00	0.90
B. Sun glare	0.45	0.95	0.50	0.95	0.70
C. Invalid timestamp	0.92	0.96	0.89	0.30	0.85
D. Shared data source	0.90	0.97	0.94	0.95	0.35

In Case D, fusion treats two products from the same EO frame as independent. A separate IR observation provides no corroboration for the track.

AI-Assisted ISR and Vulnerabilities of Intelligent Perception

Student tasks and deliverable

1. Score each track	Compute τ for A–D. Accept when $\tau \geq 0.80$. Send all other tracks for review.
2. Explain the decisions	Explain why glare can leave model confidence high. Identify which accepted tracks have suspect evidence.
3. Apply minimums	Keep $\tau \geq 0.80$ and require $S \geq 0.50$, $D \geq 0.60$, and $C \geq 0.50$. Record every failed requirement.
4. Test the weights	For D, shift 0.10 from β to η and recompute τ. Compare reweighting with a context minimum and propose a source-provenance check.

Submit

A table with each score, both decisions, and all review reasons. Add a short paragraph explaining why source independence matters in sensor fusion.

AI-Assisted ISR and Vulnerabilities of Intelligent Perception

Scores and track decisions

Case	Score τ	Score only	With minimums	Reason for review
A	0.9160	Accept	Accept	All requirements met
B	0.6975	Review	Review	Score and sensor quality
C	0.8205	Accept	Review	Metadata validity
D	0.8505	Accept	Review	Context consistency

Case D with β = 0.15 and η = 0.25 scores 0.7885 and goes to review.

Glare can obscure vehicle features while leaving patterns the detector associates with vehicles. A context minimum blocks weak corroboration regardless of the total score. Track source-frame IDs to detect shared inputs before fusion.

Mission Resilience for ISR UAV Operations

Mission scenario

In simulation, a UAV surveys a flood-affected area and sends imagery to a ground station to help assess road access.

Command and control	The operator uses a command-and-control (C2) link to supervise the aircraft.
Imagery delivery	A separate payload data link sends imagery. Onboard storage is available if that link degrades.
Mission decisions	At each checkpoint, use five quality scores to assess whether the ISR task can continue.

The thresholds and actions in this exercise apply to the simulated mission.

Mission Resilience for ISR UAV Operations

Mission resilience score

$$R_{\mathrm{ISR}} = w_c Q_{C2} + w_n Q_{NAV} + w_p Q_{PAY} + w_d Q_{DATA} + w_s Q_{SAFE}, \quad \sum_i w_i = 1, \tag{13.3}$$

Each quality score ranges from 0 to 1. Higher values indicate better condition.

Quality input	Weight	Meaning
C2	0.25	Availability and reliability of command and control
NAV	0.25	Reliability of the navigation estimate
PAY	0.20	Ability of the sensor payload to collect usable data
DATA	0.15	Timely delivery of ISR products to the ground station
SAFE	0.15	Margin relative to simulated flight safety limits

Mission Resilience for ISR UAV Operations

Simulated mission checkpoints

Checkpoint	C2	NAV	PAY	DATA	SAFE
A. Nominal	0.95	0.95	0.90	0.95	1.00
B. ISR downlink loss	0.90	0.92	0.88	0.40	0.95
C. Navigation degradation	0.96	0.45	0.95	0.95	0.90
D. C2 and safety degrade	0.55	0.93	0.95	0.90	0.55

In B, the imagery downlink degrades while the C2 link remains usable.

In C, navigation quality drops even though the other inputs remain high.

Mission Resilience for ISR UAV Operations

Decision rules and student tasks

1. Score-only rule Calculate the resilience score for A–D. Mark “continue” if the score is at least 0.80 and “pause” otherwise.

2. Ordered mission rules Apply the first matching rule in the table. Record the trigger for each checkpoint.

Priority	Condition	Simulated action
1	C2 < 0.70 or SAFE < 0.80	End ISR and invoke the contingency mode
2	NAV < 0.60 or score < 0.80	Pause ISR and reassess
3	DATA < 0.60	Continue limited ISR and buffer imagery
4	All other cases	Continue ISR as planned

Recalculate C with NAV = 0.76 and all other inputs fixed. Submit scores, both decisions, and the triggers for A–D and the recovery check.

Mission Resilience for ISR UAV Operations

Checkpoint scores and decisions

Case	Score	Score only	Ordered rules	Trigger
A	0.9475	Continue	Continue ISR	All requirements met
B	0.8335	Continue	Limited ISR	DATA below minimum
C	0.8200	Continue	Pause ISR	NAV below minimum
D	0.7775	Pause	End ISR	C2 and SAFE below minimum

B continues in limited mode. The UAV buffers imagery while C2 remains usable.

C passes the overall score threshold but pauses because NAV falls below its minimum.

Mission Resilience for ISR UAV Operations

Worked example and recovery

Case C at the degraded checkpoint

$$0.25 \times 0.96 + 0.25 \times 0.45 + 0.20 \times 0.95 + 0.15 \times 0.95 + 0.15 \times 0.90$$

Score = 0.8200. Pause ISR because NAV = 0.45 is below 0.60.

Case C after navigation improves

$$0.8200 + 0.25 \times (0.76 - 0.45) = 0.8975$$

The recovered score meets the overall threshold and all input minimums. The simulated mission can resume ISR as planned.